



# Varicose Veins

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**Varicose veins are abnormally enlarged superficial veins in the legs.**

- Varicose veins may cause your legs to ache, itch, and feel tired.
- Doctors can detect varicose veins by examining the skin.
- Surgery or injection therapy can remove varicose veins, but new ones often form.

(See also [Overview of the Venous System](#).)

## Causes of Varicose Veins

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The precise cause of varicose veins is unknown, but the main problem is probably a weakness in the walls of superficial veins (veins located just under the skin). This weakness may be inherited. Over time, the weakness causes the veins to lose their elasticity. Veins stretch and become longer and wider. To fit in the same space that they occupied before they stretched, the elongated veins become convoluted. They may appear as a snakelike bulge beneath the skin.

In people with varicose veins, the widening of the veins causes the valve flaps (cusps or leaflets) to separate (see figure [One-Way Valves in the Veins](#)). When the person stands, the blood is pulled backward by gravity and is not stopped because the valve flaps are separated. Thus, blood flows backward, rapidly filling the veins and causing the thin-walled, convoluted veins to enlarge even more. Some of the connecting veins, which normally allow blood to flow only from the superficial veins into the deep veins, also enlarge. If they enlarge, their valve flaps also separate. Consequently, blood flows backward into the superficial veins when the muscles squeeze the deep veins, causing the superficial veins to stretch further.

Women may be more likely to develop varicose veins than men, and their initial occurrence may be during pregnancy. Many people with varicose veins also have [spider veins](#), which are enlarged capillaries.

## Blood Vessels

VIDEO



## Varicose Veins of the Leg

IMAGE



Varicose veins are enlarged and often twisted veins. On this person, they are more prominent on the left leg.

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## Symptoms of Varicose Veins

Varicose veins commonly ache and cause a sensation of tiredness in the legs. However, many people, even some with very large veins, have no pain.

The lower part of the leg and ankle may itch, especially if the leg is warm after a person has been wearing socks or stockings. Itching can lead to scratching and can cause redness or a rash, which is often incorrectly attributed to dry skin. The pain is sometimes worse when varicose veins are developing than when they are fully stretched.

Only a small percentage of people with varicose veins have complications, such as [dermatitis](#), [superficial venous thrombosis](#), inflammation of the veins (phlebitis), or bleeding. People with varicose veins can also develop [chronic venous insufficiency](#).

Phlebitis may occur spontaneously or result from an injury. Although usually painful, phlebitis that occurs with varicose veins is rarely harmful.

[Dermatitis](#) causes a red, scaling, itchy rash or a brown area that may be more apparent on light skin than dark skin, usually on the inside of the leg above the ankle. Scratching or a minor injury, particularly resulting from shaving, can cause bleeding or development of a painful ulcer that does not heal. Ulcers may also bleed.

## Diagnosis of Varicose Veins

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- Doctor's evaluation
- Sometimes Doppler ultrasound

Varicose veins can usually be seen bulging under the skin, particularly when people are standing.

Malfunction of the deep veins is suggested by changes in the skin or by swollen ankles. The ankles swell because fluid accumulates in the tissue under the skin—a condition called [edema](#). Varicose veins alone do not cause edema.

When people are having symptoms, ultrasound is used to identify abnormalities in the vein's valves, as well as both obstruction and backwards flow. (see [Chronic Venous Insufficiency and Post-thrombotic Syndrome](#)).

## Treatment of Varicose Veins

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- Support hose
- Sometimes, injection or laser therapy
- Sometimes surgery

Although individual varicose veins can be removed or eliminated by injection therapy or surgery, the disorder cannot be cured. Thus, treatment mainly relieves symptoms, improves appearance, and prevents complications. Elevating the legs—by lying down or using a footstool when sitting—relieves the symptoms of varicose veins but does not prevent new varicose veins from forming. Usually, varicose veins that appear during pregnancy largely subside during the 2 or 3 weeks after delivery. During this time, they should not be treated.

**Elastic stockings** (support hose) compress the veins and prevent them from stretching and hurting. People who do not want surgery or injection therapy or who have a medical condition that prevents them from having these treatments may choose to wear elastic stockings.

### Laser or radiofrequency ablation

Ablation with lasers or radiofrequency energy can be used for the treatment of varicose veins. This treatment uses a highly focused, continuous stream of energy to cut or destroy tissue. This treatment is sometimes used when a person desires cosmetic improvement.

### Injection therapy (sclerotherapy).

Injection therapy seals the veins so that blood can no longer flow through them. A solution, such as [sodium tetradecyl sulfate](#), is injected into the vein to irritate it and produce a blood clot (thrombus). In essence, this procedure produces a harmless kind of [superficial venous thrombosis](#). Healing of the blood clot leads to formation of scar tissue, which blocks the vein. However, the blood clot may dissolve instead of becoming scar tissue, and the varicose vein then reopens. Also, new varicose veins often develop.

Sclerotherapy techniques also use special bandaging that reduces the size of the blood clot by compressing the diameter of the injected vein. A smaller blood clot is more likely to form scar tissue, as desired. A further advantage of this technique is that adequate compression virtually eliminates the pain that usually accompanies the vein irritation caused by the technique.

Although injection therapy is more time-consuming than surgery, it has several advantages:

- Anesthesia is not necessary.
- New varicose veins can be treated as they develop.
- People can go about their normal daily activities between treatments.

### Surgery ("vein stripping").

Surgery is no longer used as a routine treatment for varicose veins. When surgery is done, it aims to remove as many of the varicose veins as possible. However, surgeons try to preserve the saphenous vein. This vein is the longest superficial vein in the body, extending from the ankle to the groin, where it joins the femoral vein (the main deep vein in the leg). Surgeons try to preserve the saphenous vein because it can be used for bypass procedures in case blockage ever develops in the heart vessels or other major body vessels. If the saphenous vein must be removed, a procedure called stripping is done. The surgeon makes two incisions, one at the groin and one at the ankle, and opens the vein at each end. A flexible wire is threaded through the entire vein and then pulled out to remove the vein.

To remove other varicose veins, the surgeon makes incisions in other areas. Because the superficial veins play a less significant role than the deep veins in returning blood to the heart, their removal does not impair circulation if the deep veins are functioning normally.

Removal of varicose veins is a lengthy procedure, so the person is usually given a general anesthetic. This procedure relieves the symptoms and prevents complications, but it leaves scars. The more

extensive the procedure, the longer the time before new varicose veins develop. However, removal of varicose veins does not eliminate the tendency to develop new varicose veins.

## More Information

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The following English-language resource may be useful. Please note that The Manual is not responsible for the content of this resource.

[Vascular Cures: Varicose veins](#)

## Drug Information for the Topic

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